



YEAR 12 ASSESSMENT TASK 3

June 23rd 2011

MATHEMATICS EXTENSION 1.

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 Minutes.
- A table of standard integrals is provided at the back of this paper.
- Start each question on a new page.
- All necessary working should be shown in every question.

QUESTION	MARK
1	
2	
3	
Total	/45

Instructions: **Answer each question on a new page**

Question 1 **(15 marks)** **Marks**

a) Find the exact value of $\cos\left(\sin^{-1}\frac{40}{41}\right)$ (2)

b) i) Find $\frac{d}{dx}\sin^{-1}\left(\frac{4x}{3}\right)$. (2)

ii) Hence calculate $\int_0^{\frac{3}{4}} (9 - 16x^2)^{\frac{-1}{2}} dx$ (2)

c) i) Given that $x^2 + 4x + 5 \equiv (x + a)^2 + b^2$, find a and b (2)

ii) Hence $\int \frac{1}{x^2 + 4x + 5} dx$ (1)

d) Differentiate $y = \cos^{-1}\sqrt{3x^2 - 1}$ (2)

e) If $f(x) = 2\sin^{-1}3x$ find

i) the domain and range of $f(x)$ (2)

ii) Sketch $f(x) = 2\sin^{-1}3x$ (2)

QUESTION 2. (16 Marks) Answer on a new page.

- a) N is the number of kangaroos in a certain population at time t years. The population size N satisfies the equation

$$\frac{dN}{dt} = -k(N - 500) \text{ for some constant } k.$$

- i) Verify that $N = 500 + Ae^{-kt}$ where A is a constant, is a solution of the equation (2)
 - ii) Initially, there are 3500 kangaroos but after 3 years there are only 3300 left. Find the value of A and the exact value of k . (2)
 - iii) Find when the number of kangaroos begins to fall below 2300. (2)
 - iv) Sketch the graph of the population size against time. (2)
- b) The speed v m/s of a particle moving in a straight line is given by $v^2 = 64 - 16x - 8x^2$ where the displacement from a fixed point O is x metres.
- i) Find an expression for the acceleration and show the motion is simple harmonic. (3)
 - ii) Between which two points is the particle oscillating? (2)
 - iii) Find the period and amplitude of the motion. (2)
 - iv) Find the maximum speed of the particle (1)

QUESTION 3. (14 Marks) Answer on a new page.

- a) If $\frac{dx}{dt} = x + 6$ and $x = -5$ when $t = 0$, find an expression for x in terms of t . (3)
- b) The volume of a cube is expanding at the constant rate of $5\text{mm}^3/\text{sec}$. At what rate is the surface area of the cube increasing when the side length of the cube is 60 centimetres. (3)
- c) i) The curve $y = x^4$ is rotated one revolution about the y axis to form a container for storing water. Calculate the volume of water that can be stored if the container is filled to a depth of h metres (3)
- ii) Water is poured into the above container at a rate of 60 ml/minute . Find the rate at which the depth is increasing when the depth is 16cm. (3)
- d) The equation of motion of a particle moving along a horizontal straight line is given by the formula $x = 3\cos\left(\frac{1}{4}t\right) + \sin\left(\frac{1}{4}t\right)$ where x is the displacement of the particle at time t seconds.
- Explain whether the particle is initially moving to the right or left and whether it is speeding up or slowing down. (2)

END OF ASSESSMENT TASK.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

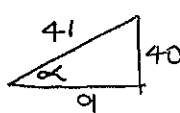
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

Question 1

a)  $\cos \alpha = \frac{9}{41}$

b) i) $\frac{1}{\sqrt{1 - (\frac{4x}{3})^2}} \times \frac{4}{3} \quad (1)$

$$= \frac{3}{\sqrt{9 - 16x^2}} \times \frac{4}{3}$$

$$= \frac{4}{\sqrt{9 - 16x^2}}$$

ii) $\frac{1}{4} \int_0^{3/4} \frac{1}{\sqrt{9 - 16x^2}} dx$

$$\frac{1}{4} \left[\sin^{-1} \left(\frac{4x}{3} \right) \right]_0^{3/4}$$

$$\frac{1}{4} \left[\sin^{-1} 1 - \sin^{-1} 0 \right]$$

$$\frac{1}{4} \times \frac{\pi}{2} = \frac{\pi}{8}$$

c) $x^2 + 4x + 5 = x^2 + 4x + 4 + 1$
 $= (x+2)^2 + 1^2$
 $a=2 \quad b=1 \quad (2)$

ii) $\int \frac{1}{(x+2)^2 + 1^2} dx = \tan^{-1}(x+2) + c \quad (1)$

d) $\frac{-1}{\sqrt{1 - (3x^2-1)}} \times \frac{1}{2} (3x^2-1)^{-1/2} \cdot 6x \quad (3x^2-1)^{1/2}$

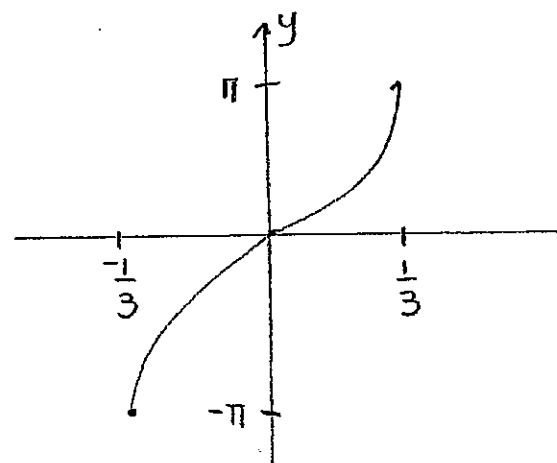
$$= \frac{-1}{\sqrt{2-3x^2}} \times \frac{3x}{\sqrt{3x^2-1}} \quad \text{or} \quad = \frac{-3x}{\sqrt{2-3x^2} \cdot \sqrt{3x^2-1}} \quad (2)$$

e) i) $-1 \leq 3x \leq 1$
 $-\frac{1}{3} \leq x \leq \frac{1}{3} \quad (1)$

$$-\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$$

$$-\pi \leq y \leq \pi \quad (1)$$

ii)



(2)

QUESTION 2

i) $N = 500 + Ae^{-kt}$

$$\frac{dN}{dt} = -kAe^{-kt}$$

$$But Ae^{-kt} = N - 500$$

$$\therefore \frac{dN}{dt} = -k(N - 500)$$

ii) $t=0 \quad N=3500$

$$3500 = 500 + A$$

$$A = 3000$$

$$3300 = 500 + 3000e^{-3k}$$

$$2800 = 3000e^{-3k}$$

$$\frac{14}{15} = e^{-3k}$$

$$\log_e\left(\frac{14}{15}\right) = -3k$$

$$k = -\frac{1}{3} \log_e\left(\frac{14}{15}\right)$$

iii) $N < 2300$

$$500 + 3000e^{-kt} < 2300$$

$$3000e^{-kt} < 1800$$

$$e^{-kt} < \frac{3}{5}$$

$$\log_e e^{-kt} < \log_e \frac{3}{5}$$

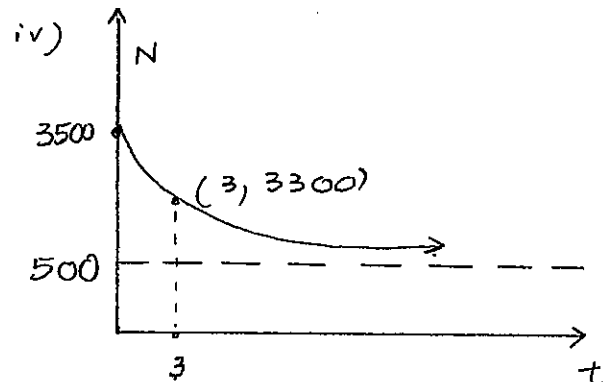
$$\frac{1}{3} \log_e\left(\frac{14}{15}\right)t < \log_e \frac{3}{5}$$

$$t < \frac{\log_e \frac{3}{5}}{\frac{1}{3} \log_e\left(\frac{14}{15}\right)}$$

$$-0.0229t < -0.5108$$

$$t > 22.3 \quad (22.21) \quad (2)$$

allow rounding.



b) $\frac{1}{2}v^2 = 32 - 8x - 4x^2$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -8 - 8x$$

$$\ddot{x} = -8(1+x)$$

$$\therefore \ddot{x} = -n^2(x)$$

$\therefore$ SHM centre $x = -1$

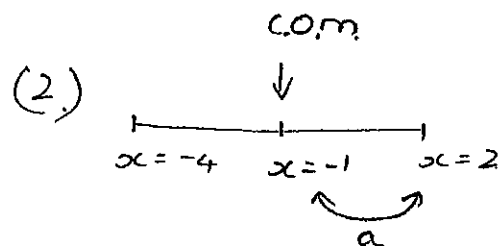
ii) let $v^2 = 0$ (at rest)

$$0 = 64 - 16x - 8x^2$$

$$0 = x^2 + 2x - 8$$

$$0 = (x+4)(x-2)$$

$$x = -4 \quad x = 2$$



iii) $T = \frac{2\pi}{n}$

$$= \frac{2\pi}{\sqrt{8}} \quad \text{or} \quad \frac{\pi}{\sqrt{2}}$$

(1)

$$a = 3$$

(1)

v) max speed at centre of motion $x = -1$

$$v^2 = 64 - 16x - 1 - 8x(-1)^2$$

$$= 64 + 16 - 8$$

$$= 72$$

$$v = \sqrt{72} \quad \text{or} \quad 6\sqrt{2} \text{ m/s.}$$

(1)

QUESTION 3

a)

$$\frac{dx}{dt} = x + 6$$

$$\frac{dt}{dx} = \frac{1}{x+6}$$

$$t = \ln(x+6) + c$$

when $t = 0$ $x = -6$

$$0 = \ln 1 + c$$

$$\therefore c = 0$$

$$t = \ln(x+6)$$

$$e^t = x + 6$$

$$x = e^t - 6$$

b) $V = x^3$ $\frac{dV}{dt} = 5$

$$\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$$

$$5 = 3x^2 \cdot \frac{dx}{dt}$$

$$\frac{5}{3x^2} = \frac{dx}{dt}$$

$$\frac{dA}{dt} = \frac{dA}{dx} \cdot \frac{dx}{dt}$$

$$= 12x \cdot \frac{5}{3x^2}$$

$$= \frac{20}{x}$$

When $x = 60 \text{ cm} = 600 \text{ mm}$

$$\frac{dA}{dt} = \frac{20}{600} = \frac{1}{30} \text{ mm}^2/\text{sec}$$

c) i) $V = \pi \int_0^h \sqrt{y} \, dy$ y axis
 $y = x^4$
 $\sqrt{y} = x^2$
 $= \pi \int_0^h y^{1/2} \, dy$
 $= \pi \left[\frac{2}{3} y^{3/2} \right]_0^h = \pi \left[\frac{2}{3} h^{3/2} \right]$
 $= \frac{2\pi}{3} h^{3/2}$

ii) $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ $\frac{dV}{dh} = \frac{3}{2} \times \frac{2}{3} \pi h^{1/2}$
 $60 = \pi \sqrt{h} \cdot \frac{dh}{dt}$ $= \pi \sqrt{h}$

$$\frac{60}{\pi \sqrt{h}} = \frac{dh}{dt}$$

$$\frac{15}{\pi} = \frac{dh}{dt}$$

d) $\dot{x} = -\frac{3}{4} \sin \frac{t}{4} + \frac{1}{4} \cos \left(\frac{t}{4} \right)$

$$\ddot{x} = -\frac{3}{16} \cos \frac{t}{4} - \frac{1}{16} \sin \left(\frac{t}{4} \right)$$

$t=0 \quad v = \frac{1}{4}, \quad \ddot{x} = -\frac{3}{16}$

$v > 0$ particle moving to right
 $\ddot{x} < 0$ slowing down